

ERP Management System Documentation

This document contains the complete overview, architecture, modules, technology stack, deployment process, and implementation details of the ERP Management System project.

1. Project Overview

The ERP Management System is a full-stack enterprise management application developed using React.js, Spring Boot, and PostgreSQL. The project manages employees, attendance, finance, inventory, recruitment, salaries, projects, products, tasks, and performance tracking in a centralized system.

2. Technology Stack

- Frontend: React.js, JavaScript, Tailwind CSS
- Backend: Spring Boot, Java
- Database: PostgreSQL / Neon Database
- ORM: Hibernate / JPA
- API Testing: REST APIs
- Version Control: Git & GitHub
- Deployment: Render + Neon

3. System Architecture



The frontend communicates with backend REST APIs developed in Spring Boot. The backend processes business logic and stores/retrieves data from PostgreSQL.

4. ERP Modules

1. Employee Management
2. Department Management
3. Attendance Tracking
4. Leave Management
5. Finance Management
6. Inventory Management
7. Recruitment System
8. Project Management
9. Task Management
10. Product Management
11. Salary Management
12. Performance Monitoring
13. User Authentication

5. Backend Structure

The backend follows layered architecture with Controllers, Services, Repositories, Entities, and Configuration packages.

Backend Entities

- Attendance
- Department
- Employee
- Finance
- Inventory
- Leave
- Performance
- Product
- Project
- Recruitment
- Salary
- Task
- User

6. Frontend Structure

The frontend is developed using React.js with reusable components, pages, services, API integrations, and responsive UI.

- Dashboard
- Employee UI
- Reports
- Authentication
- API Service Layer
- Responsive Design

7. Database Design

The project uses PostgreSQL database with separate tables for each ERP module. Spring Data JPA handles ORM mapping automatically.

8. REST API Flow

The frontend sends HTTP requests to backend APIs. The backend validates requests, processes business logic, and interacts with the database.

9. Deployment Process

14. Push frontend and backend code to GitHub
15. Create Neon PostgreSQL cloud database
16. Update application.properties with Neon credentials
17. Deploy backend using Render
18. Deploy frontend using Render/Vercel
19. Connect frontend with deployed backend APIs

10. Key Features

- Centralized ERP system
- Cloud database support
- Secure REST APIs
- Modular architecture
- Responsive UI
- Scalable backend
- Role-based structure

11. Future Enhancements

- JWT Authentication
- Role-based Access Control
- Analytics Dashboard
- Email Notifications
- AI-based Reporting
- Mobile Application

12. Resume Project Description

Developed a full-stack ERP Management System using React.js, Spring Boot, and PostgreSQL to manage enterprise operations including employee management, attendance, finance, inventory, recruitment, projects, and reporting.

13. Conclusion

The ERP Management System successfully integrates multiple enterprise operations into a single platform. The project demonstrates strong full-stack development skills, database management, REST API development, and deployment knowledge.